These are the realities of this geography. So, when we design EVs for mass adoption, we have to ensure that they are built to match the needs. There is no point in creating something that cannot compete with existing vehicles, both in terms of price and performance. Failure in either of these areas can slow down the adoption of EVs, because the technology should be such that apples can be compared to apples.

## Perspective of driver/owner

There are various people whose livelihood depends on the performance of the vehicle, such as 3-wheeler and 4-wheeler drivers. They need a vehicle that is sturdy and lasts long. It is also important to ensure that the vehicle, once charged, can be driven whole day without any worry. A well-sized battery pack can provide operational assurance and reduce down time.

A commercial vehicle, in particular, should be able to carry heavy loads. And financing options are very critical, so we need to provide those tailored financing and leasing options that suit the consumers.

Driver behavior can significantly affect the operating expenditure and longevity of the battery. Batteries are the most expensive part in an EV, and unprofessional driving patterns can degrade the battery faster. Therefore, it is important to have the right kind of technology stacks in an EV, not just efficient drive trains. We need software that can constantly monitor battery dynamics and help in increasing efficiency. The API can be provided so that data can be extracted by the fleets or customers to ensure better driving behaviour.

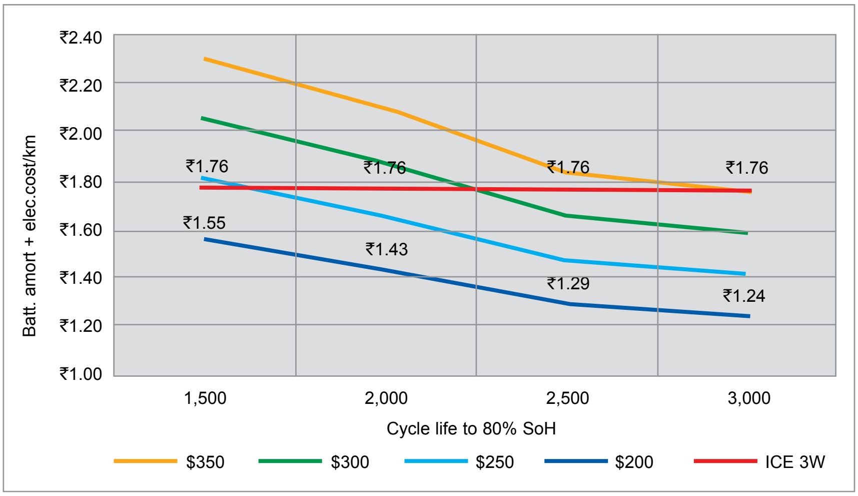


*Fig. 2: EV3W - Cycle Life vs Energy cost/km for different battery costs*

## Expected trends

EV penetration will occur primarily in commercial fleets with 2- and 3-wheelers, and small commercial 4-wheelers. Total cost of ownership and resale value are affected by energy efficiency and battery's state of health, and both of these are dictated by driver's behavior. Driver behavior can impact the vehicle range by 20% or more, and battery life by a similar percentage.

In future, fleet operations will need telematic platforms that offer integrated EV services. There is a scope for technology to improve battery-life prediction and real-time interventions to encourage good driving. The telematic platform is expected to monitor vehicle diagnostics, driver behaviour, battery usage, trip planning and scheduling, charging infrastructure planning, usage and scheduling, etc. All that is very important to ensure predictive support, so that downtime can be reduced.

While autonomous vehicles (AVs) are unlikely to emerge soon in the Indian market, AV technologies will find application in off-road cases as well as for energy efficiency and safety uses.

To cater to all these trends and requirements for adoption of EVs, innovative solutions are critical. Automotive OEMs and startups must make sure that they create solutions and designs that can compete in both price and performance with existing technologies. EFY

*The article is based on a presentation given by Dr Amitabh Saran, Founder & CEO, Altigreen Propulsion Labs, at MOVES 2021.*